

SATHVANI AVC

System Architecture & Tech Stack — Visual Flow

Artificial Vocal Cord | Version 1.0 | August 2026 | Confidential

1. System Overview

Sathvani AVC is a distributed **edge-cloud architecture** that restores natural, personalized speech for people who have lost their voice (laryngectomy, vocal-cord paralysis, ALS). Physiological signals captured by a wearable mask are streamed to the cloud, reconstructed into phonemes by a CNN-BiLSTM-Attention model, converted into words via dictionary + language-model re-ranking, and synthesized into personalized speech using FastSpeech2 + HiFi-GAN.

2. End-to-End Architecture Flow

The system is organized into four logical layers. Data flows strictly top-to-bottom.

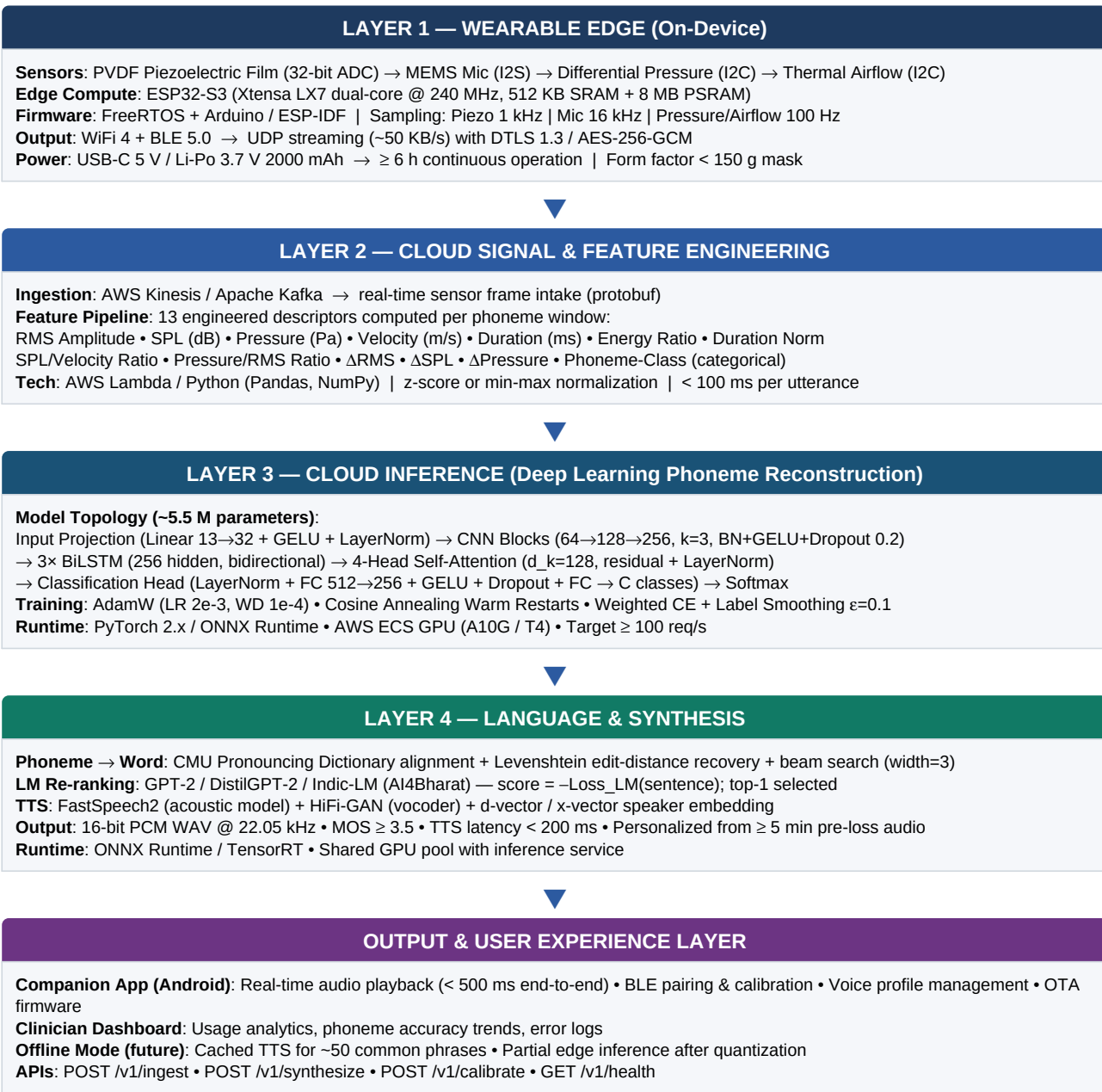


Figure 1. Complete data path from physiological sensors to personalized speech output.

3. Connected Tech Stack by Layer

Every component is mapped to the concrete technologies specified in the Technical Requirements Document.

Sensor — Vibration	PVDF Piezo Film + external 32-bit ADC	~50 pC/N, 0.1 Hz–20 kHz; primary signal (~60-70% SHAP)
Sensor — Acoustic	MEMS Mic (SPH0645 or equiv.) via I2S	16-bit, 16 kHz, SNR 64 dB
Sensor — Pressure	Differential Pressure (MPXV7002 or equiv.) I2C	±2 kPa, 2.5% FS accuracy
Sensor — Airflow	Thermal Mass (FS1015 or equiv.) I2C	0–30 m/s, ±3% accuracy
Edge MCU	ESP32-S3 (Xtensa LX7 dual-core)	240 MHz, WiFi 4 + BLE 5.0, FreeRTOS
Streaming Protocol	UDP over WiFi + application-layer reliability	Header (seq, ts, mask) + payload + CRC16; DTLS 1.3
Cloud Ingestion	AWS Kinesis / Apache Kafka	Real-time streams; 50 KB/s × 2,300 devices (Y1)
Feature Engineering	AWS Lambda + Python (Pandas, NumPy)	13-descriptor vector; < 100 ms latency
Inference Service	AWS ECS (GPU) + PyTorch / ONNX Runtime	A10G instance; ≥ 100 req/s target
Language Model	GPT-2 / DistilGPT-2 / Indic-LM (HuggingFace)	Re-ranking for sentence plausibility
TTS Engine	FastSpeech2 + HiFi-GAN	Non-autoregressive mel + GAN waveform; < 200 ms
Speaker Embedding	d-vector / x-vector (256-dim)	Personalization from 5-min pre-loss audio
API Gateway	AWS API Gateway / Cloudflare	Rate limiting, auth, routing; 10 K req/day (Y1)
Database	PostgreSQL + Redis	User profiles, voice models, logs; 100 GB
Object Storage	AWS S3 (encrypted)	Audio, model artifacts, datasets; 5 TB (Y1)
Monitoring	Prometheus + Grafana + CloudWatch	Latency, accuracy, drift, health dashboards
CI/CD & IaC	GitHub Actions + Terraform	Full pipeline automation
Mobile App	Android (Kotlin / Java) + BLE 5.0	Pairing, playback, calibration, OTA
Security (Transit)	TLS 1.3 / DTLS 1.3	All device-cloud and app-cloud channels
Security (Rest)	AES-256-GCM	Patient audio, sensor data, voice models
AuthN / AuthZ	OAuth 2.0 + JWT (RS256) + RBAC	Patient / Clinician / Admin / Researcher roles
Compliance	DPDP Act 2023 • HIPAA • GDPR • ISO 13485 • IEC 62304	Consent, localization, BAA, QMS, risk management

Table 1. Complete technology mapping from sensors to compliance frameworks.

4. Model Architecture Snapshot

The core phoneme reconstruction network is a lightweight CNN-BiLSTM-Attention stack designed for cloud inference with a clear path to edge quantization (INT8 via ONNX / TFLite Micro on ESP32-S3 PSRAM, later ESP32-P4 / Ethos-U55).

Input	13 engineered descriptors	(B, T, 13)	—
Projection	Linear(13→32) + GELU + LayerNorm	(B, T, 32)	~448
CNN Blocks	3× Conv1d (32→64→128→256, k=3) + BN + GELU + Dropout(0.2)	(B, 256, T)	~129 K
BiLSTM	3 layers, 256 hidden, bidirectional	(B, T, 512)	~4.2 M
Self-Attention	4 heads, d_k=128, residual + LayerNorm	(B, T, 512)	~1.05 M
Classification	LN + FC(512→256) + GELU + Drop + FC → C	(B, T, C)	~132 K
Total	Lightweight — suitable for cloud + future edge		~5.5 M

Table 2. CNN-BiLSTM-Attention model topology (from TRD §5.1).

5. Feature Importance (Research Baseline)

SHAP values from an auxiliary XGBoost model on the 13-descriptor space (research dataset). Pressure, velocity and ratio features currently show near-zero contribution — a critical monitoring metric for real-patient data collection.

1	RMS Amplitude	0.4134	Dominant energy predictor
2	Phoneme-Class	0.2671	Articulation group prior
3	Δ RMS	0.1842	Local energy transition
4	Duration	0.1336	Phoneme length discrimination
5	SPL	0.1058	Sound pressure level
6	Δ SPL	0.0677	Local SPL transition

Table 3. Top SHAP-ranked features on research baseline (TRD §5.3).

6. Critical Path & Edge Roadmap

Critical Path to Commercial Launch:

1. Sensor signal validation on patients (Phase 1–2) → gates AI model reliability
2. AI model v2 training on real patient data (Phase 3) → gates clinical efficacy claim
3. 10-patient pilot completion at AIIMS Bibinagar (Phase 3) → gates regulatory submission
4. CDSCO manufacturing license (Phase 4) → gates commercial sales in India
5. FDA 510(k) / De Novo clearance (Phase 4) → gates US market entry

Edge Quantization Roadmap:

- **Current:** Full inference in cloud; ESP32-S3 does acquisition + streaming only.
- **2027:** INT8 quantization of CNN-BiLSTM (ONNX + TFLite Micro); partial phoneme classification on edge (top-3 candidates); ≤ 2 MB model size.
- **2028:** Evaluate ESP32-P4 or ARM Cortex-M55 + Ethos-U55 NPU for full edge inference; < 100 ms latency for 50 common phrases; cloud reserved for personalization and analytics.

Source documents: AVC Technical Requirements Document (TRD) v1.0, Product Requirements Document (PRD) v1.0, Implementation Plan v1.0 — August 2026. All figures and mappings derived directly from these specifications.

— End of Architecture & Tech Stack Document —